

ANIL PRADHAN

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AI & Backend Engineer specializing in autonomous multi-agent architectures, computer vision pipelines, and resilient backend systems with Python, FastAPI, LangGraph, OpenCV, and YOLO11n.

EDUCATION

Institute of Technical Education & Research (ITER), SOA University

Bachelor of Technology (B.Tech), Computer Science & Engineering

2024 – 2028

CGPA: 8.93

Nalanda Vidya Mandir Higher Secondary School, Berhampur, Odisha

Class XII (Higher Secondary) — Physics, Chemistry, Mathematics with IT

2022 – 2024

Score: 85%

Odisha Adarsha Vidyalaya (OAV), Digapahandi, Ganjam, Odisha

Class X (CBSE Board)

2017 – 2022

Score: 78.8%

TECHNICAL PROJECTS

VisionForge AI — AI Hardware Forensic Inspector for Electronics QA | [GitHub](#)

Aug 2026 – Present

AI/ML & Pipeline Engineer | Team: Disha & Anil

- Built an autonomous inspection pipeline using **LangGraph state machines** to run OCR, seal-matching, and YOLO11n detection in parallel, with **Groq and Gemini Vision** models giving the final pass/fail verdict in under 5s.
- Trained and fine-tuned a custom **YOLO11n model on 4,448 hardware images** to accurately detect and locate electronic components as small as **12×12 px**.
- Developed an anomaly detection approach (**Anomaly Max-Pooling**) using **OCR and FAISS** to improve detection of critical hardware defects, with **203 automated tests**.
- Implemented **real-time inspection updates using Server-Sent Events (SSE)**, allowing operators to track pipeline progress and receive inspection results instantly.

Tech: Python · FastAPI · LangGraph · YOLO11n · OpenCV · FAISS · Groq LPU · Gemini 3.5 Flash

TaskPilot AI — AI Chief of Staff for Software Engineers | [GitHub](#) | [Live Demo](#)

Jul 2026 – Present

Integration Lead | Team IdeaForge-E

- Built the **orchestrator** for a 6-stage AI pipeline, coordinating task collection, extraction, deduplication, quality checks, prioritization, and scheduling across **5 data sources**.
- Implemented the **pipeline lifecycle and state management** to handle stage execution and recovery, completing the full workflow in around **14–18 seconds**.
- Built a **reliable LLM integration** with fallback mechanisms, allowing the system to continue working during API failures and rate limits.
- Implemented **real-time P1 task injection** through the AI copilot, automatically triggering a pipeline re-run and moving urgent incidents to the **#1 priority**.

Tech: Python · FastAPI · Multi-Agent Systems · SQLAlchemy · SQLite (WAL) · Groq · React 18

AI Career Mentor — AI Career Coach & Mock Interviewer for Developers | [GitHub](#) | [Live Demo](#)

Feb 2026 – Aug 2026

Solo Developer

- Built a **production-grade FastAPI backend** with REST APIs, JWT authentication, request validation, and rate limiting for a full-stack career platform.
- Designed the **PostgreSQL + Redis data layer** for persistent user data, caching, session management, and feature usage limits.
- Set up **Docker-based deployment and GitHub Actions CI/CD** with **113 automated tests**, Prometheus metrics, and Sentry monitoring for reliable production operation.

Tech: FastAPI · PostgreSQL · Redis · WebSockets · Docker · GitHub Actions · SQLAlchemy

TRAINING

Agentic AI Training Program — CogniForce.AI & ITER, SOA University

2026 – Present

- Hands-on training in **Agentic AI, LLMs, tokenization, prompt engineering, RAG, tool calling, MCP**, and multi-agent orchestration.
- Built autonomous agents using **OpenCode, Multica, AnythingLLM, Claude Code, MCP**; integrated multi-cloud LLM APIs across **Groq, OpenRouter, Google AI Studio, and NVIDIA Build**.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL

AI & Computer Vision: LangGraph, Multi-Agent Systems, RAG, Prompt Engineering, LLM Integration, OpenCV, YOLO11n, OCR, Anomaly Detection (SSIM)

Backend & Databases: FastAPI, REST APIs, SQLAlchemy, JWT/RBAC, PostgreSQL, Redis, ChromaDB, SQLite

Core CS Concepts: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks

Developer Tools: Git/GitHub, Docker, Linux Commands, Postman

ACHIEVEMENTS & CERTIFICATIONS

- Dell FutureMinds AI Hackathon 2026 — National Grand Finalist:** Top 6 nationwide out of 8,000+ participants; awarded Certificate of Excellence & pitched live at Dell Bengaluru HQ.
- FlowFinance × Techfest IIT Bombay — Round 2:** Developed GTM strategy and acquisition framework for AI fintech platform.
- 4× AI Hackathon Builder:** Deployed AI applications in Microsoft AI Dev Days, Amazon Nova, and Google Solution Challenge.
- 100+ Day DSA Streak:** Continuous daily problem solving on GeeksforGeeks 160 Days of Code.
- Certifications:** IBM AI Fundamentals · Cisco Python 2 · Wells Fargo SWE (Forage) · HackerRank (DSA, SQL, Java, Python).